Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Conducting polymers - examples, types of doping

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Conducting polymers - examples, types of doping

1 Conducting polymers - examples, types of doping

Study focus: Conducting polymers - examples, types of doping. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000146 · EV0000147

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module seven on functional materials that is polymers and today's class we are going to talk about conducting polymers, a brief introduction, the types of dopamine, the applications of conducting polymers.

Key details

- Polymers are generally known for their insulating property because the organic molecules of which they are composed have no free electrons to carry current.
- An organic polymer with highly delocalized pi electron system and having an electrical conductance of the order of an electrical conductor.
- Such a polymer was referred to as a conducting polymer.
- We know that copper has very copper metal has very high conductivity of the order of 10 raised to the power of eight sements per meter, whereas Teflon has conductance of 10 to the power of minus 8 siemens per meter.
- And if you have doped polyaniline, it is found to have 10 to the power of 5 sements per meter.
- These are also known as synthetic metals.
- Heger, McDermott, and Shirakawa won the Nobel Prize for Chemistry for the du discovery of conducting polymers.
- The answer is definitely no.
- poly thiophene, polyaniline, and here we can see polyphenylin vinyline.
- what we can see they have a linear structure.
- All those polymers which we listed just now they have a linear structure and here's I've taken just two examples.

- Which should be there in their structure, and that is extensive conjugation in its polymer backbone.
- What would that mean?
- That they should have alternate.
- single and double bond.

Example or application from the lecture

- So let us talk about the development of conducting polymers because just recently in the year 2000.
- And let us have a look at some examples of such polymers.
- For example, polyacetylene, then we have polyphenylin, then this is polypyro.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 05:00–09:59

When what is doping what are we doing here?

Key details

- We are introducing a charged chemical species, we find that the conduction increases manifold.
- There could be oxidative doping, there could be reductive doping, or there could also be proton doping.
- The pi backbone of the polymer is partially oxidized with a suitable oxidizing agent.
- It is going to take away electron from the polymer backbone.
- And now we understand why we chose a.
- Pi system where because we know that pi electrons they can easily be removed from the they are more mobile or as we say as compared to the sigma bar sigma electrons, sigma bond the electrons which are forming the sigma bond.
- See these plants are much more free to move.
- So therefore they can be easily removed from the system.
- That's why we chose the pi electron system here.
- This creates a positively charged site on the polymer backbone and now these become the current carriers for conduction.
- They are going to remove an electron from the pi system, hence we get a positive charge over here.
- In order to compensate for the positive charge, we know that you have learned during different mechanisms that these electrons will move to the next bond, these will move to the next bond, hence the electrons are moving in one direction and the hole is moving a positive charge or the whole is moving in another direction just like it happens in.
- P type doping and what is moving along the chain when a potential is applied is the electron or this positive charge, they are moving in opposite direction, and that is how the polymer becomes conducting when we do oxidative doping.
- The mechanism how it is happening.

- So when a potential is applied, so again because these electrons can be delocalized, so these electrons now move in the opposite direction, and therefore this negative charge moves in one direction, and again the hole is going to move in the direction.
- the opposite direction.
- So because of this pi electron system, the alternate extensive conjugation which we have in this polymer backbone, this type of movement is possible.
- The third type of doping is known as protonic acid doping.
- And what is here the current carrying charge species that is the positive and the negative species are again created and that is with the help of a protonic acid.
- The current carrying charge species that the that is the positive and the negative species are created by protonation of imine nitrogen.

Example or application from the lecture

- And that is how now we have negatively charged sites which are moving and hence we call it as N type of doping and this can be done with reducing agents like for example sodium naphtalide in tetrahydrofural.
- Now example best example would be doping of polyaniline and in polyaniline.
- Let us just have a look at what the structures look like.

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Core explanation · 10:00–14:59

Partial oxidation is done with the help of ammonium peroxy disulfate.

Key details

- So you will have alternate this will be alternated along the entire polyanyline change chain and now what we are going to do the next step is a protonation that is done with H Cl that is one molar H Cl.
- H plus i that's why it's called protonic acid doping.
- This gets the plus charge and now it is balanced with the help of Cl minus.
- Cl minus so this is now becoming a plus n h plus n h plus and now this is balanced with the help of Cl minus and here also Cl minus because we added H Cl.
- So this is how it got doped, and now because of the delocalized pi electron which are available here, we can have these charges moving along the along the chain.
- Emeraldine salt and this is green in color and this structure is what is conducting polyanine so you can write here conducting polyanum.
- So as such, in this case we don't find it to be conducting the first structure.
- When it undergoes partial oxidation, what we get is known as the amoraldine base.
- This is also not conducting.
- It's only after protonation when the MRL wing salt is formed, then.
- This is this is found to be green in color, and this is the structure of conducting polyanuline.
- And how did we make it conducting?

- We know now that polymer are conducting in nature some of these polymers and they can be made conducting by doping.
- But when we talk of electrodes, we are talking of electronic conduction.
- Now what we saw in conducting polymers is electronic conduction because the electron could move through the polymer chain.
- So therefore they will be used as electrode material and because they are lightweight we expect a very high power to weight ratio.
- Then they can be used as conductive tracts on PCBs normally we see copper being printed there.
- Then they can be used as sensors.
- Like humidity sensor, gas sensor, radiation sensor, biosensors, etc.
- Then electrochromic display window, specially polyaniline, is known to have different colours in different atmospheres.
- Now the color of each of these is different and therefore it can be used in electrochromic display windows to tell us what atmosphere is there inside the system which is which is being studied.
- Then in information storage devices as film membranes for gas separations in light emitting diodes and they are important materials in microelectronics applications, then electro catalysis fuel cell electrodes and reinforced composites etc.
- And now in the next class we are talking about the molecular weight of polymers.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Conducting polymers - examples, types of doping transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Conducting polymers - examples, types of doping transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Conducting polymers - examples, types of doping transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/048_Conducting_polymers_-_examples_types_of_doping__64decc463863d8a0.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00039	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00040	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00041	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

C Revision and retrieval practice